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(54) **MECHANICAL STOP ARRANGEMENT FOR A BOREHOLE TUBULAR TO LOCATE AND RESTRAIN A BOREHOLE TOOL TO AN OUTSIDE DIAMETER OF THE TUBULAR, METHOD, AND SYSTEM**

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See application file for complete search history.

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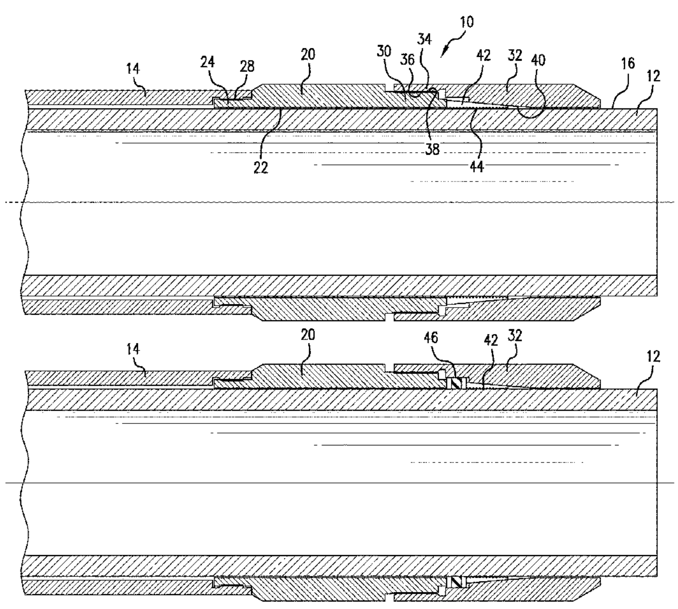
(57) **ABSTRACT**

A mechanical stop arrangement includes a first coupler having a tool connection segment, and an arrangement connection segment, a second coupler having a first connection segment connectable with the arrangement connection section, and a frustoconical inside diameter surface, and a wedge having a radially outside surface angle that is complementary to the frustoconical inside diameter surface. A method for connecting a borehole tool to an outside surface of a tubular, including connecting the borehole tool to a first coupler either before or after disposing the coupler on the tubular, disposing the first coupler on the tubular, disposing a wedge on the tubular, disposing a second coupler on the tubular, interconnecting the first coupler and the second coupler, and jamming the wedge against the tubular with the interconnecting. A wellbore system, including a borehole, a string in the borehole, and an arrangement, disposed within or as a part of the string.

22 Claims, 8 Drawing Sheets



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E21B 43/10	(2006.01)
E21B 43/12	(2006.01)

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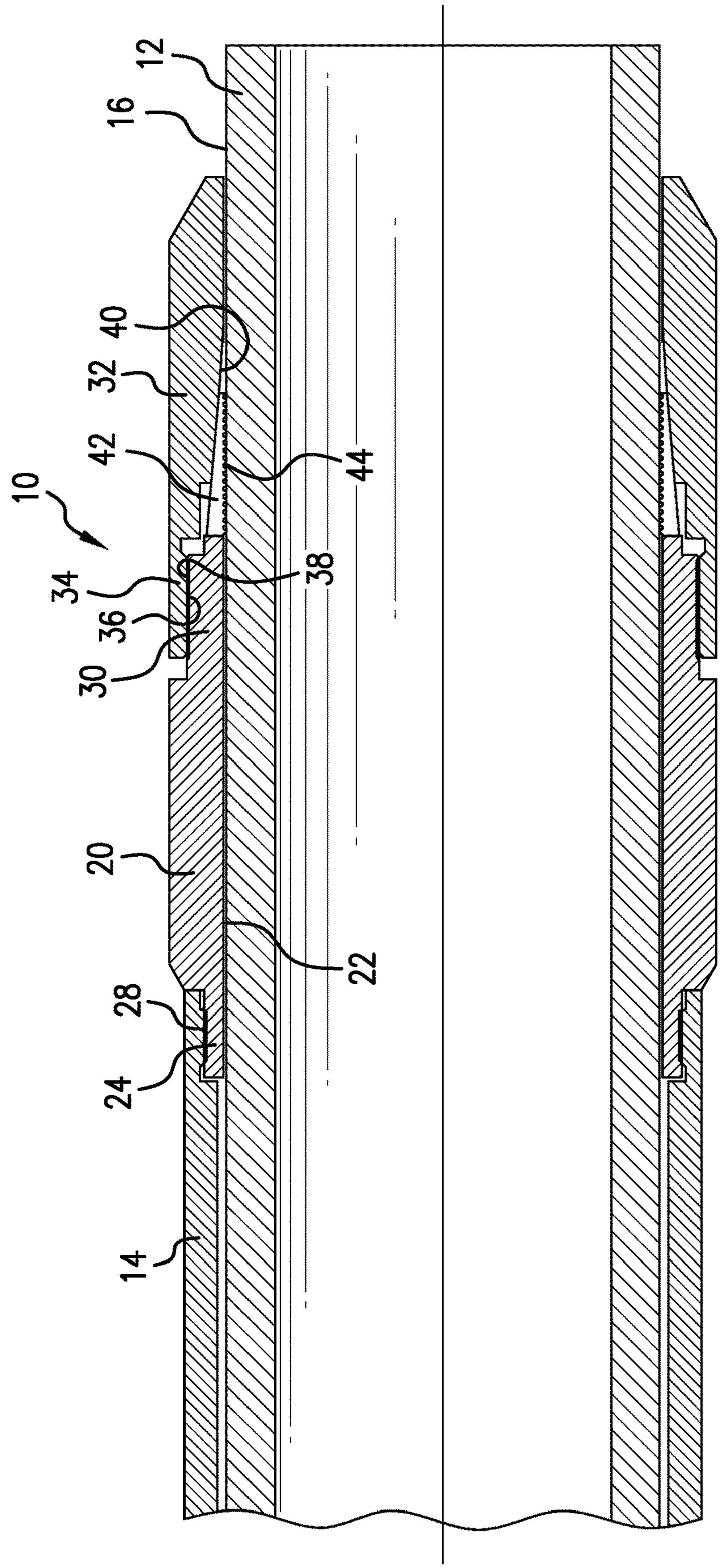

FIG. 1

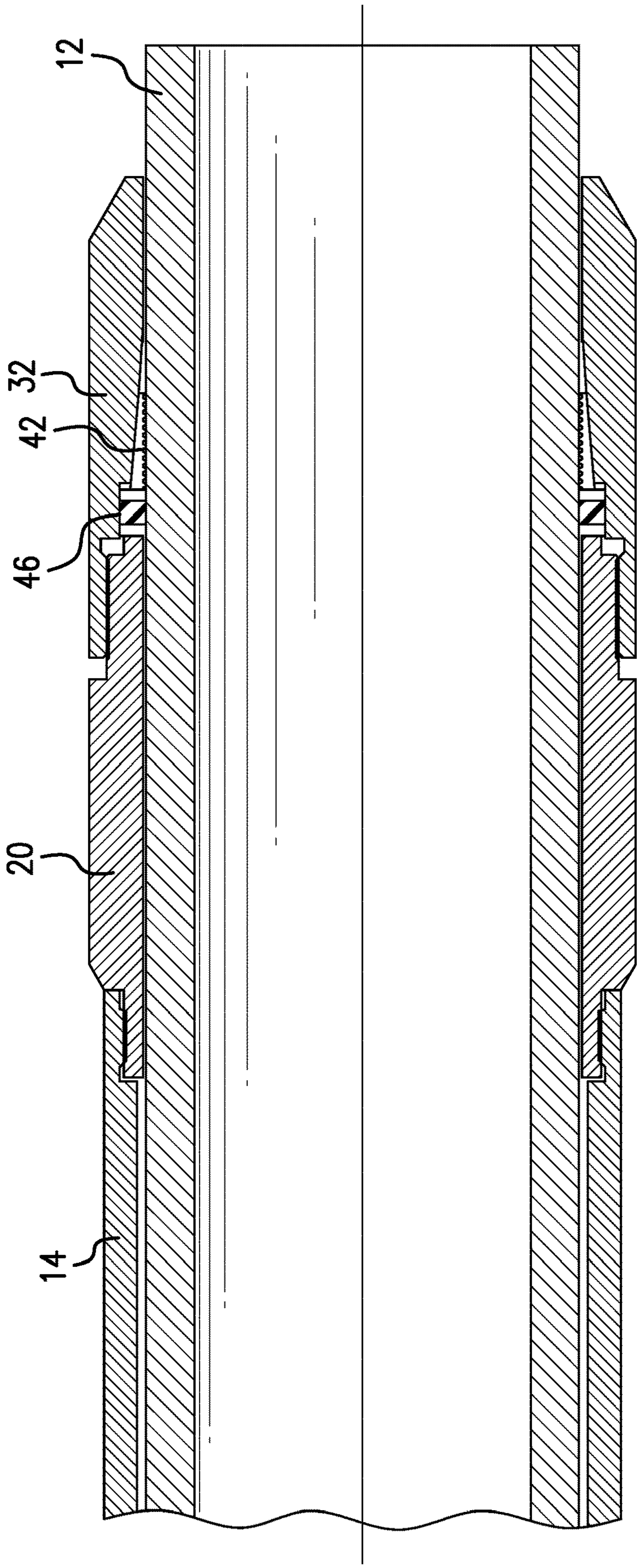

FIG. 2

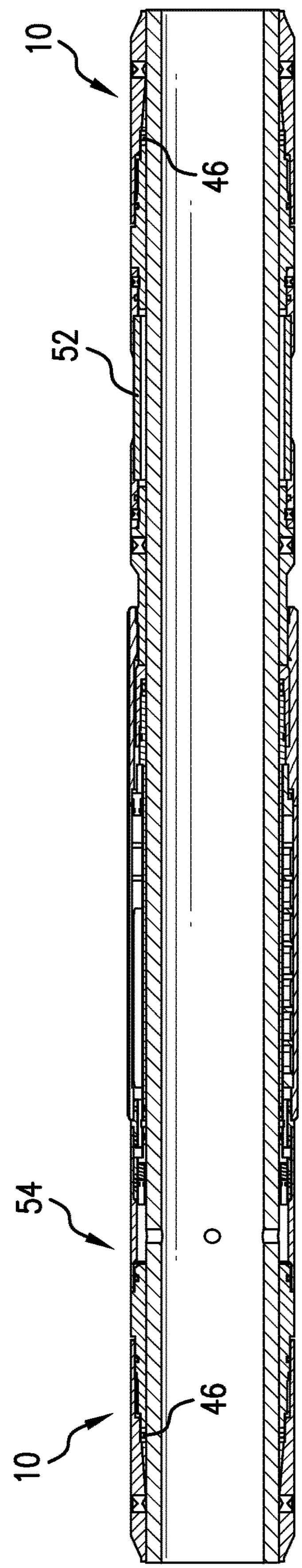

FIG.3

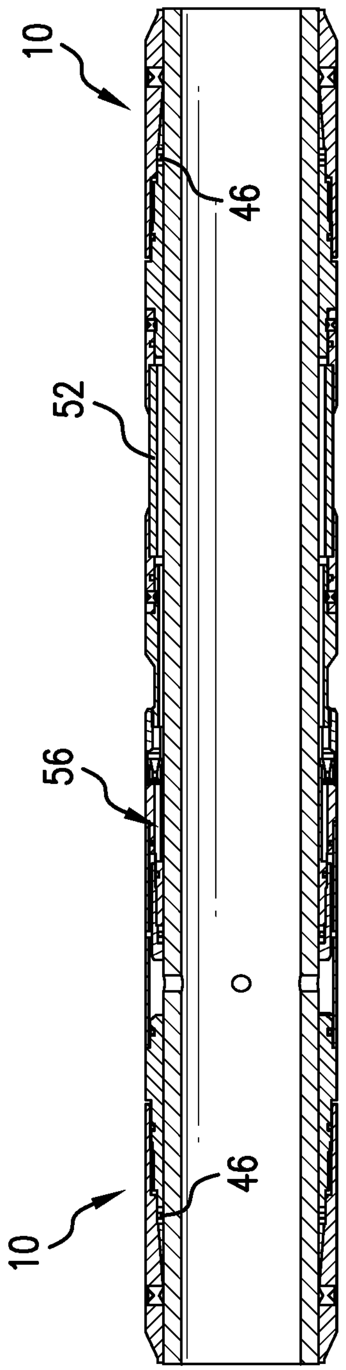

FIG.4

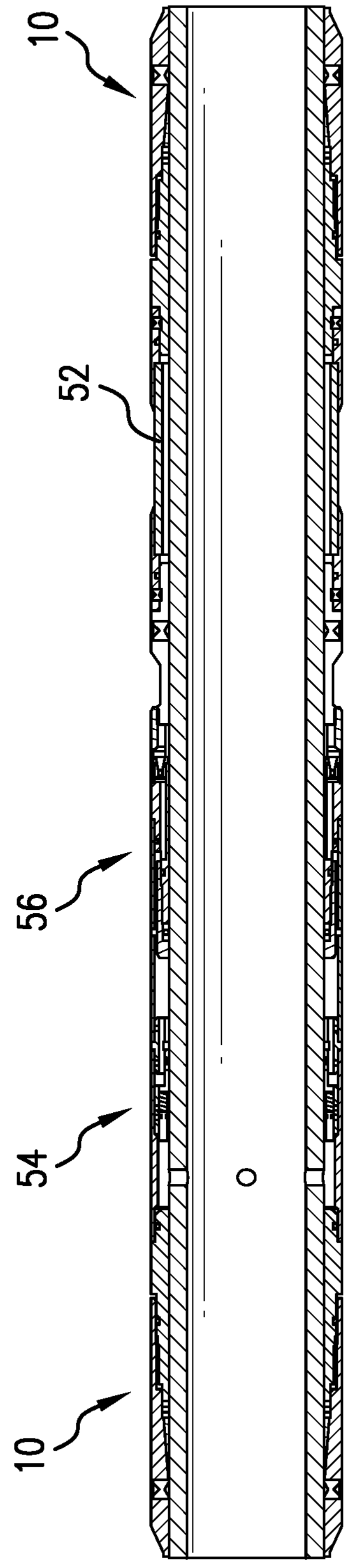

FIG.5

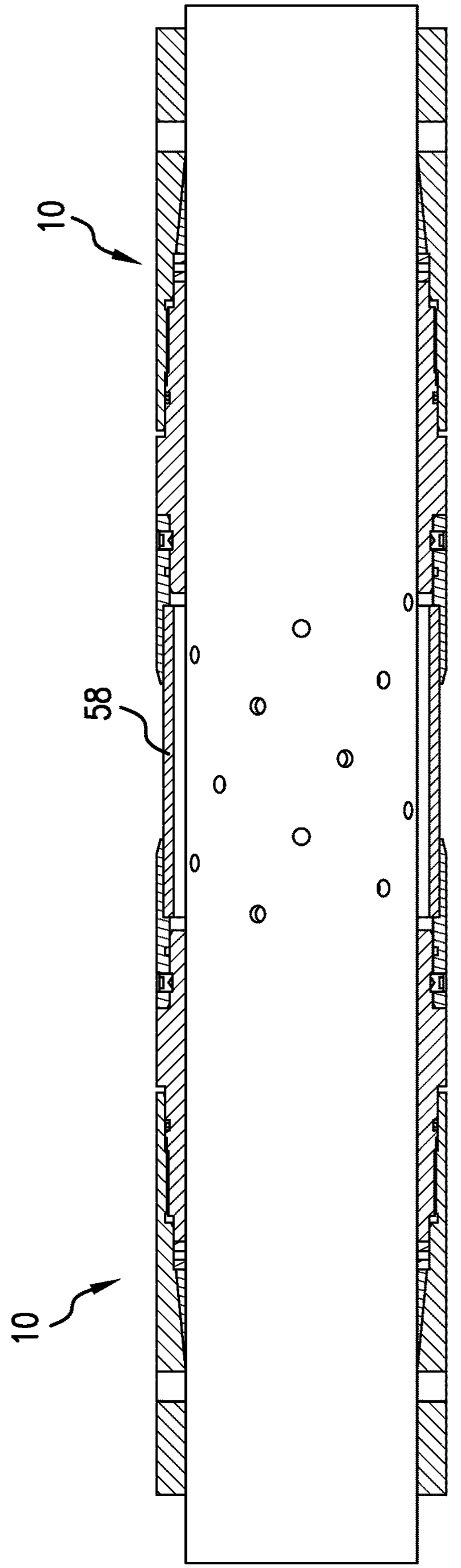

FIG.6

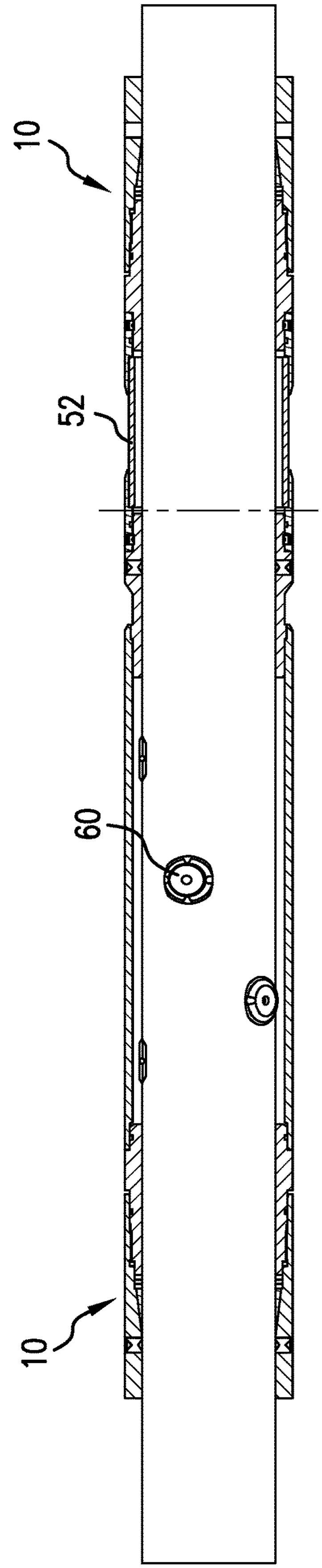

FIG. 7

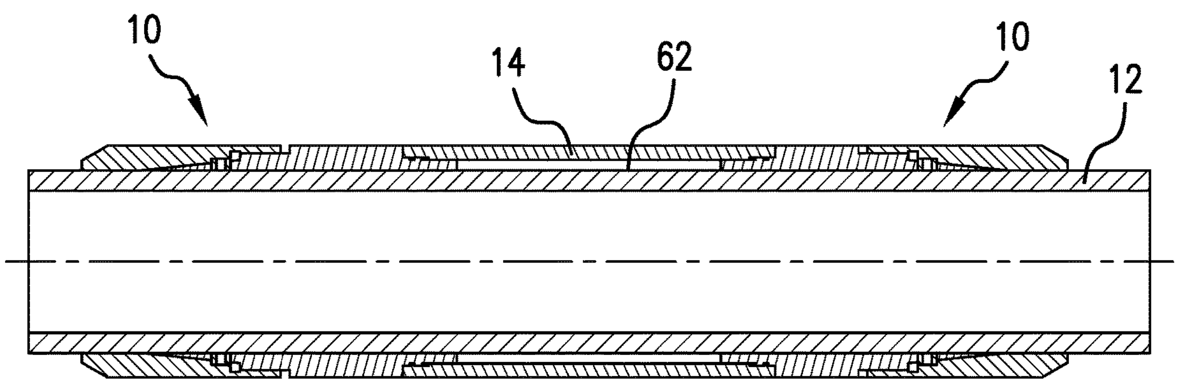

FIG.8

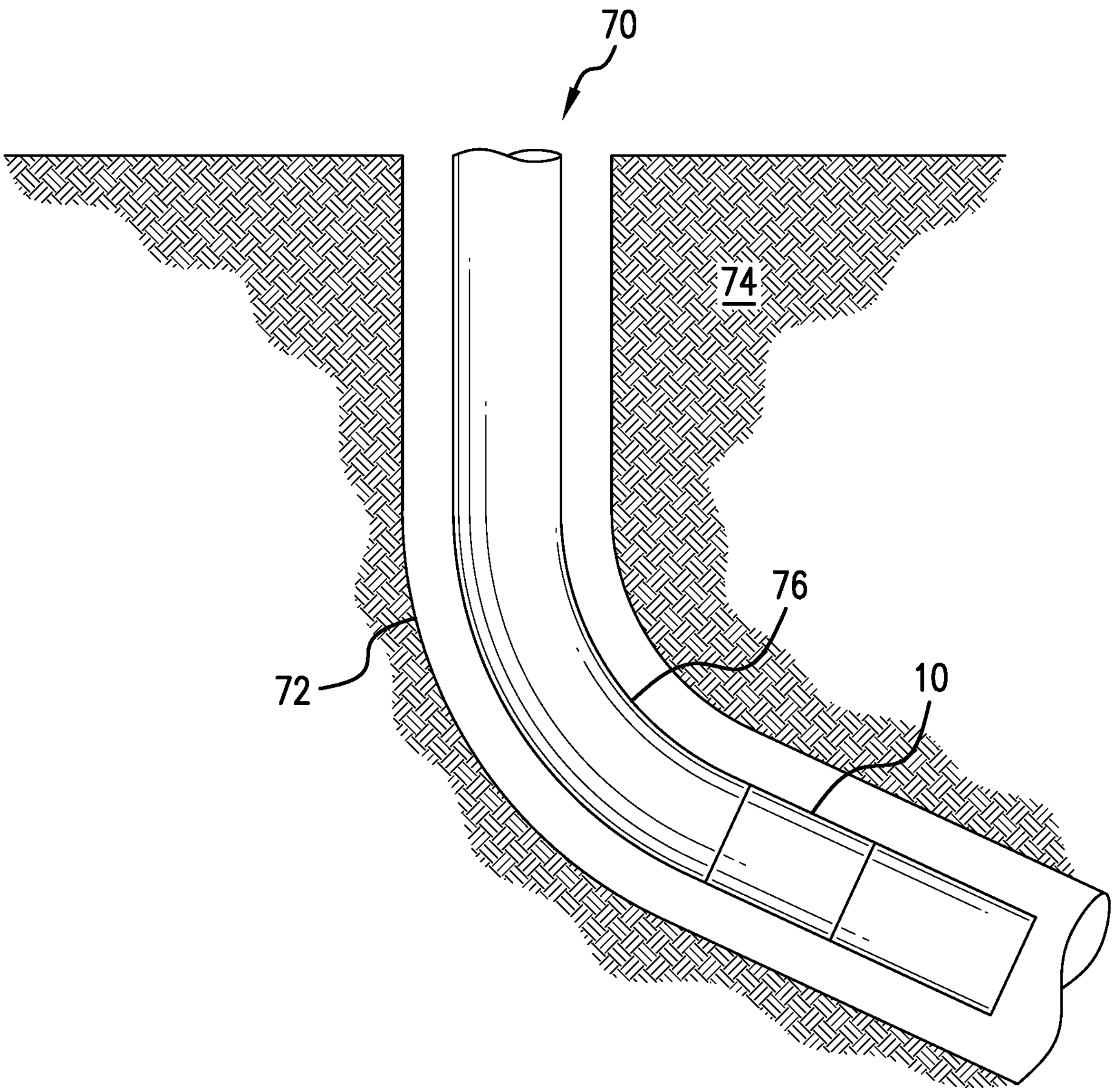

FIG.9

MECHANICAL STOP ARRANGEMENT FOR A BOREHOLE TUBULAR TO LOCATE AND RESTRAIN A BOREHOLE TOOL TO AN OUTSIDE DIAMETER OF THE TUBULAR, METHOD, AND SYSTEM

BACKGROUND

In the resource recovery and fluid sequestration industries, tools that are secured to an outside diameter of a tubular are often welded thereto. Alternatively, bracketry is known that uses set screws or clamping arrangements where two halves of a bracket are bolted together. Welding requires highly skilled labor and the brackets that are clamped onto tubulars are large and cumbersome. The only answer for operators has been to incur the expense and delay associated with sending components to a shop specifically set up to weld. The art would well receive alternatives that reduce cost and make local modifications for specific well operations easy to do on site.

SUMMARY

An embodiment of a mechanical stop arrangement for a borehole tubular to locate and restrain a borehole tool to an outside diameter of the tubular, including a first coupler dimensioned and configured to be disposed about the borehole tubular and having a tool connection segment configured to connect with a downhole tool, and an arrangement connection segment, a second coupler dimensioned and configured to be disposed about the borehole tubular and having a first connection segment connectable with the arrangement connection section, and a frustoconical inside diameter surface, and a wedge having a radially outside surface angle that is complementary to the frustoconical inside diameter surface.

An embodiment of a method for connecting a borehole tool to an outside surface of a tubular, including connecting the borehole tool to a first coupler either before or after disposing the coupler on the tubular, disposing the first coupler on the tubular, disposing a wedge on the tubular, disposing a second coupler on the tubular, interconnecting the first coupler and the second coupler, and jamming the wedge against the tubular with the interconnecting.

An embodiment of a wellbore system, including a borehole in a subsurface formation, a string in the borehole, and an arrangement, disposed within or as a part of the string.

BRIEF DESCRIPTION OF THE DRAWINGS

The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

FIG. **1** is a schematic view of a mechanical stop arrangement for a borehole tubular to locate and restrain a borehole tool to an outside diameter of the tubular;

FIG. **2** is a schematic view of a mechanical stop and seal arrangement for a borehole tubular to locate and restrain a borehole tool to an outside diameter of the tubular;

FIG. **3** is a section view of a system employing the stop of FIG. **1**;

FIG. **4** is a section view of another system employing the stop of FIG. **1**;

FIG. **5** is a section view of another system employing the stop of FIG. **1**;

FIG. **6** is a section view of another system employing the stop of FIG. **1**;

FIG. **7** is a section view of another system employing the stop of FIG. **1**;

FIG. **8** is a section view of another system employing the stop of FIG. **1**; and

FIG. **9** is a view of a borehole system including is the mechanical stop arrangement as disclosed herein.

DETAILED DESCRIPTION

A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

Referring to FIG. **1**, a mechanical stop arrangement **10** for a borehole tubular **12** to locate and restrain a borehole tool **14** to only an outside diameter surface **16** of a separate borehole tubular **12** that extends past each of two opposing first and second longitudinal ends of the arrangement **10**. The arrangement **10** facilitates attachment of borehole tools to an outside surface of tubular members such as drill pipe, coiled tubing, etc. without modification of the underlying tubular. The arrangement **10** is usable in any location on the tubular **12** and can be assembled in the field using relatively unskilled labor. The arrangement **10** allows for in the field and on the fly modification of borehole tools being selected and combined for use all while avoiding the need for welding or other laborious and skilled worker intensive treatments. The arrangement **10** includes a first coupler **20** that is configured with an inside diameter **22** that is a clearance fit for a target type of tubular **12**. The clearance fit is of tight tolerance of not more than 0.125 inch. First coupler **20** includes a tool connection segment **24** configured to connect with the downhole tool **14**. The connection may be a threaded connection **28** or may employ radially directed fastening features (such as screws, pins, etc.). the tool **14** is illustrated schematically and is intended to represent inflow control devices such as screens, valves, etc., or a liner hanger, a packer, electronics (such as sensors, gauges, batteries, processors, memory, etc.). It will be noted that the first coupler **20** at least, may have a radial dimension that is larger than a radial dimension of the tool **14** and hence may provide impact protection for the tool **14**. At an opposite end of first coupler **20**, is an arrangement connection segment **30**.

A second coupler **32** is attachable to the first coupler **20** at a first connection segment **34**. In an embodiment, the first connection segment **34** exhibits a thread **36** that is engagable with a complementary thread **38** on the arrangement connection segment. Second coupler **32** defines a frustoconical inside surface **40** having an angle measured from a longitudinal axis of the arrangement **10** of about 3 to about 15 degrees. In one embodiment, the angle is about 3 to about 5 degrees.

Disposed radially inwardly of the second coupler **32**, when assembled, is a wedge **42**. Wedge **42** may be fully annular, part annular, a split ring, or a series of one or more slip-type members arranged annularly. When in use, the wedge **42** is trapped between the frustoconical surface **40** and the tubular surface **16** to create a load bearing interface that will not slide. In some embodiments, the wedge may be smooth while in others, the wedge **42** may be configured with a texture at least on a tubular interface surface **44**. Textures include serrations, teeth, wickers, knurls, thread, roughness, etc. and any of the surfaces of the wedge **42** may be hard-faced by such processes as heat treating, carburizing, Nitriding, etc.

It is further noted that the wedge **42** may be configured to create a seal with the surface **40** and the surface **16** using a metal-to-metal seal, a coating seal (elastomeric, polymeric, soft metal, etc.). Sealing as such may be obtained regardless of the type of wedge disclosed above but some types of sealing are better associated with some types of wedge such as an annular ring wedge is best for a metal to metal seal.

In another embodiment of the arrangement **10**, referring to FIG. **2**, a dedicated seal **46** is added. The components are labeled with the same numerals as they are substantially identical but for additional length left for the seal **46**. Seal **46** may be a packing element, an O-ring, a bonded seal, or other sealing configuration. Seal **46** is energized into sealing contact with the surface **16** of the tubular **12**, the second coupler **32** and the first coupler **20** during threading of the first and second couplers together. Accordingly, while the wedge **42** is being driven into the surface **16**, the seal is also making sealing contact therewith. This arrangement allows use with tools **14** that require pressure differentials to be applied to actuate the tools.

Referring to FIG. **3**, one system is illustrated with an arrangement **10** on either end and supporting a tool **14**. In this case the tool **14** includes a screen **52** and a pressure actuated valve **54** such as an MTV™ pressure actuated valve available from Baker Hughes, Houston, Texas. Due to the seals **46** in this system, pressure can be managed to actuate the tool **14** properly.

Referring to FIG. **4**, another exemplary system is illustrated. This is similar to FIG. **3** but includes a nozzle **56**.

Referring to FIG. **5**, yet another exemplary system is illustrated that includes both the pressure actuated valve and the nozzle. The various configurations of FIGS. **3-5** illustrate the effective modularity and modifiability of the ultimate system in the field at point of use. Due to the arrangement **10**, the various tools **14** may be disposed upon the tubular **12** and secured thereon using one or more of the arrangements **10**. These systems may also be reconfigured and or repaired as well due to the fact that the arrangement **10** may be unscrewed to remove the tools **14**.

FIG. **6** illustrates another exemplary system that employs a sand screen **58** as the tool **14**.

FIG. **7** illustrates another exemplary system that employs a sand screen **52** and a Bernoulli type valve **60** as the tool **14**.

FIG. **8** illustrates another exemplary system where a chamber **62** is created to house, for example electronic components in a protected environment. In this case the tool **14** is a sleeve that bridges between two arrangements **10** and creates the chamber **62** annularly between the tool **14** and the tubular **12**. The protection afforded may be merely from impact without sealing or may be also from wellbore fluids using the sealing described hereinabove. Sensitive componentry may be protected anywhere along the tubular using the mechanical stop arrangements **10** as described herein.

Referring to FIG. **9**, a borehole system **70** is illustrated. The system **70** comprises a borehole **72** in a subsurface formation **74**. A string **76** is disposed within the borehole **72**. An arrangement **10** as disclosed herein is disposed within or as a part of the string **76**.

Set forth below are some embodiments of the foregoing disclosure:

Embodiment 1: A mechanical stop arrangement for a borehole tubular to locate and restrain a borehole tool to an outside diameter of the tubular, including a first coupler dimensioned and configured to be disposed about the borehole tubular and having a tool connection segment configured to connect with a downhole tool, and an arrangement connection segment, a second coupler dimensioned and configured to be disposed about the borehole tubular and having a first connection segment connectable with the arrangement connection section, and a frustoconical inside diameter surface, and a wedge having a radially outside surface angle that is complementary to the frustoconical inside diameter surface.

Embodiment 2: The arrangement as in any prior embodiment, further comprising a seal disposable between the first coupler and second coupler and configured and dimensioned to create a seal against the tubular, when the arrangement is in use.

Embodiment 3: The arrangement as in any prior embodiment, wherein the seal is in contact with the wedge, the second coupler and the first coupler and subject to compression by the arrangement.

Embodiment 4: The arrangement as in any prior embodiment, wherein the wedge also creates a seal.

Embodiment 5: The arrangement as in any prior embodiment, wherein the seal is metal to metal.

Embodiment 6: The arrangement as in any prior embodiment, wherein the wedge includes a sealing material coating.

Embodiment 7: The arrangement as in any prior embodiment, wherein the wedge is a ring.

Embodiment 8: The arrangement as in any prior embodiment, wherein the wedge is a split ring.

Embodiment 9: The arrangement as in any prior embodiment, wherein the wedge is a plurality of wedges arranged about the tubular when in use.

Embodiment 10: The arrangement as in any prior embodiment, wherein the wedge includes a surface texture.

Embodiment 11: The arrangement as in any prior embodiment, wherein the surface texture is teeth.

Embodiment 12: The arrangement as in any prior embodiment, wherein the surface angle is in a range of about 3 degrees to about 15 degrees from a longitudinal axis of the arrangement.

Embodiment 13: The arrangement as in any prior embodiment, wherein the surface angle is in a range of about 3 degrees to about 5 degrees from a longitudinal axis of the arrangement.

Embodiment 14: The arrangement as in any prior embodiment, wherein the borehole tool is an inflow control tool.

Embodiment 15: The arrangement as in any prior embodiment, wherein the borehole tool is a liner hanger.

Embodiment 16: The arrangement as in any prior embodiment, wherein the borehole tool is a packer.

Embodiment 17: The arrangement as in any prior embodiment, wherein the borehole tool is electronics.

Embodiment 18: The arrangement as in any prior embodiment, wherein the borehole tool creates a chamber.

Embodiment 19: The arrangement as in any prior embodiment, wherein the chamber is sealed from wellbore fluids.

Embodiment 20: A method for connecting a borehole tool to an outside surface of a tubular, including connecting the borehole tool to a first coupler either before or after disposing the coupler on the tubular, disposing the first coupler on the tubular, disposing a wedge on the tubular, disposing a second coupler on the tubular, interconnecting the first coupler and the second coupler, and jamming the wedge against the tubular with the interconnecting.

Embodiment 21: The method as in any prior embodiment, further comprising energizing a seal disposed between the first and second couplers to create a seal against the tubular.

Embodiment 22: A wellbore system, including a borehole in a subsurface formation, a string in the borehole, and an arrangement as in any prior embodiment, disposed within or as a part of the string.

The use of the terms "a" and "an" and "the" and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Further, it should be noted that the terms "first," "second," and the like herein do not denote any order, quantity, or importance, but rather are used to distinguish one element from another. The terms "about", "substantially" and "generally" are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, "about" and/or "substantially" and/or "generally" can include a range of ±8% of a given value.

The teachings of the present disclosure may be used in a variety of well operations. These operations may involve using one or more treatment agents to treat a formation, the fluids resident in a formation, a borehole, and/or equipment in the borehole, such as production tubing. The treatment agents may be in the form of liquids, gases, solids, semi-solids, and mixtures thereof. Illustrative treatment agents include, but are not limited to, fracturing fluids, acids, steam, water, brine, anti-corrosion agents, cement, permeability modifiers, drilling muds, emulsifiers, demulsifiers, tracers, flow improvers etc. Illustrative well operations include, but are not limited to, hydraulic fracturing, stimulation, tracer injection, cleaning, acidizing, steam injection, water flooding, cementing, etc.

While the invention has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited.

What is claimed is:

1. A mechanical stop arrangement for attachment to a separate borehole tubular in any location along the separate borehole tubular to locate and restrain a borehole tool to an outside diameter of the separate borehole tubular, the separate borehole tubular extending past each of two longitudinal ends of the arrangement, comprising:

- a first coupler dimensioned and configured to be disposed about the separate borehole tubular and having a clearance fit around the separate borehole tubular, the first coupler having:
  - a tool connection segment configured to connect with a downhole tool; and
  - an arrangement connection segment;
- a second coupler dimensioned and configured to be disposed about the separate borehole tubular and having:
  - a first connection segment connectable with the arrangement connection section; and
  - a frustoconical inside diameter surface; and
- a wedge having a radially outside surface angle that is complementary to the frustoconical inside diameter surface.

2. The arrangement as claimed in claim **1**, further comprising a seal disposable between the first coupler and second coupler and configured and dimensioned to create a seal against the tubular, when the arrangement is in use.

3. The arrangement as claimed in claim **2**, wherein the seal is in contact with the wedge, the second coupler and the first coupler and subject to compression by the arrangement.

4. The arrangement as claimed in claim **1**, wherein the wedge also creates a seal.

5. The arrangement as claimed in claim **4**, wherein the seal is metal to metal.

6. The arrangement as claimed in claim **4**, wherein the wedge includes a sealing material coating.

7. The arrangement as claimed in claim **1**, wherein the wedge is a ring.

8. The arrangement as claimed in claim **1**, wherein the wedge is a split ring.

9. The arrangement as claimed in claim **1**, wherein the wedge is a plurality of wedges arranged about the tubular when in use.

10. The arrangement as claimed in claim **1**, wherein the wedge includes a surface texture.

11. The arrangement as claimed in claim **10**, wherein the surface texture is teeth.

12. The arrangement as claimed in claim **1**, wherein the surface angle is in a range of about 3 degrees to about 15 degrees from a longitudinal axis of the arrangement.

13. The arrangement as claimed in claim **1**, wherein the surface angle is in a range of about 3 degrees to about 5 degrees from a longitudinal axis of the arrangement.

14. The arrangement as claimed in claim **1**, wherein the borehole tool is an inflow control tool.

15. The arrangement as claimed in claim **1**, wherein the borehole tool is a liner hanger.

16. The arrangement as claimed in claim **1**, wherein the borehole tool is a packer.

17. The arrangement as claimed in claim **1**, wherein the borehole tool is electronics.

18. The arrangement as claimed in claim **1** wherein the borehole tool creates a chamber.

19. The arrangement as claimed in claim **18** wherein the chamber is sealed from wellbore fluids.

20. A wellbore system, comprising:

- a borehole in a subsurface formation;
- a string in the borehole; and
- an arrangement as claimed in claim **1**, disposed within or as a part of the string.

21. A method for connecting a borehole tool to an outside surface of a separate tubular, comprising:

- connecting the borehole tool to a first coupler either before or after disposing the coupler on only an outside surface of the separate tubular;
- disposing the first coupler on the separate tubular;
- disposing a wedge on the separate tubular;
- disposing a second coupler on the separate tubular;
- interconnecting the first coupler and the second coupler; and
- jamming the wedge against the separate tubular with the interconnecting.

22. The method as claimed in claim **21**, further comprising energizing a seal disposed between the first and second couplers to create a seal against the tubular.

* * * * *